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Lepton Isolation and Signal versus Background Classification in the $W \to \ell\nu$ Channel with ATLAS Open Data at $\sqrt{s} = 13\,\mathbf{TeV}$

A thesis submitted in partial fulfilment of the requirements for the degree of BS Physics

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Dedication

To our beloved parents,
whose endless prayers, sacrifices, and encouragement
made this journey possible,

and to our teachers,
who lit the path of knowledge before us.

Declaration of Originality

We hereby declare that this thesis, entitled *"Lepton Isolation and Signal versus Background Classification in the $W \to \ell\nu$ Channel with ATLAS Open Data at $\sqrt{s} = 13$ TeV"*, is the result of our own work carried out under the supervision of Dr. Asif Ilyas, Department of Physics. No part of this thesis has been previously submitted for any other degree or qualification at this or any other institution. All sources of information, data, and literature used have been duly acknowledged and referenced. The ATLAS Open Data used in this study are publicly released by the ATLAS Collaboration for educational purposes.

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Abstract

A measurement of the charged-current Drell–Yan process $W \to \ell\nu$ ($\ell = e, \mu$) is presented using the ATLAS Open Data 1-lepton dataset at $\sqrt{s} = 13\,\text{TeV}$. The data were recorded at the Large Hadron Collider (LHC) and correspond to an integrated luminosity of $10\,\text{fb}^{-1}$. Events are selected with exactly one isolated, high-transverse-momentum lepton and large missing transverse energy arising from the neutrino. Three variables related to lepton isolation, the relative track isolation p_T^{cone30}/p_T, the relative calorimeter isolation E_T^{cone20}/p_T, and E_T^{miss}, are studied as a means to separate prompt leptons originating from W decays versus non-prompt or fake leptons originating from multijet production. The Jacobian peak of the transverse mass m_T distribution around the W boson mass is reconstructed and compared to Monte Carlo simulations for signal (W+jets) and backgrounds (Z+jets, $t\bar{t}$, single top, and diboson production). In addition to the cut-based selection, a Boosted Decision Tree (BDT) is trained on kinematic and isolation variables to discriminate signal and background, reaching an area under the ROC curve of approximately 0.992. Data and simulation are found to agree well, and the W boson signal is clearly observed above the background prediction.

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Chapter 1

Introduction

1.1 The Large Hadron Collider and the ATLAS Experiment

The Large Hadron Collider (LHC) at CERN is the world's largest and highest energy particle accelerator. During Run 2 (2015–2018), the LHC collided proton beams at a centre-of-mass energy of $\sqrt{s} = 13\,\text{TeV}$. ATLAS [1] is a general-purpose detector with close to 4π coverage. Its main components are an inner tracking detector inside a 2 T solenoidal magnetic field, electromagnetic and hadronic calorimeters, and a muon spectrometer with toroidal magnets. These subsystems enable accurate reconstruction of electrons, muons, jets, and missing transverse momentum.

1.2 Motivation

The W boson, mediator of the weak interaction and key element of the Standard Model (SM), decays to leptons with nearly 100% branching fraction. The signature for $W \to \ell\nu$ is one of the cleanest in hadron collider experiments: a single high-p_T isolated lepton with missing transverse energy. This channel can be used to:

- validate the Standard Model description of electroweak (EWK) processes at 13 TeV;
- provide benchmarks for lepton identification/isolation and missing-energy reconstruction;
- provide an important background estimate for many searches for new physics ($W' \to \ell\nu$, SUSY, dark matter, etc.).

1.3 Objectives of the Thesis

The objectives of this work are:

1. to understand and apply lepton isolation criteria to discriminate prompt leptons from non-prompt and fake leptons;
2. to reconstruct the transverse mass m_T of the W boson candidates and to observe the Jacobian peak;

3. to compare ATLAS Open Data with Monte Carlo predictions for signal and background processes;
4. to train a Boosted Decision Tree classifier for signal versus background separation and compare its performance with a cut-based selection.

1.4 Thesis Outline

The theoretical framework of the Standard Model, W boson production and decay, and multivariate classification is reviewed in Chapter 2. Chapter 3 is devoted to the dataset, event selection, and analysis methodology. Chapter 4 presents and discusses the results. Chapter 5 summarises the conclusions. The full analysis code is provided in Appendix A.

Chapter 2

Theoretical Framework

2.1 The Standard Model of Particle Physics

The Standard Model (SM) is a quantum field theory based on the gauge group

$$SU(3)_C \times SU(2)_L \times U(1)_Y, \tag{2.1}$$

describing the strong, weak, and electromagnetic interactions. Matter is built up from three generations of quarks and leptons. The Higgs mechanism [2] spontaneously breaks the electroweak sector $SU(2)_L \times U(1)_Y$ down to $U(1)_{\mathrm{EM}}$, giving mass to the $W^{\pm}$ and Z bosons:

$$m_W = \frac{1}{2}\, g\, v, \qquad m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}\, v, \tag{2.2}$$

where $v \approx 246\,\mathrm{GeV}$ is the Higgs vacuum expectation value. The measured W mass is $m_W = 80.377 \pm 0.012\,\mathrm{GeV}$ with a total width $\Gamma_W = 2.085 \pm 0.042\,\mathrm{GeV}$ [3].

2.2 W Boson Production at the LHC

In hadron colliders, W bosons are mainly produced by the charged-current Drell–Yan process [4]:

$$u\bar{d} \to W^+, \qquad d\bar{u} \to W^-. \tag{2.3}$$

The cross section factorises into parton distribution functions (PDFs) $f_q(x, Q^2)$ and the partonic cross section $\hat{\sigma}$:

$$\sigma_{pp\to W} = \sum_{q,q'} \int dx_1\, dx_2\; f_q(x_1, Q^2)\, f_{\bar{q}'}(x_2, Q^2)\, \hat{\sigma}_{q\bar{q}'\to W}. \tag{2.4}$$

Since there are two valence u quarks and only one valence d quark in the proton, there will be more W^+ than W^- produced in pp collisions, resulting in a charge asymmetry. For $\sqrt{s} = 13\,\mathrm{TeV}$, the total W production cross section multiplied by the leptonic branching ratio is around 20 nb for each lepton flavour [5].

2.3 Leptonic Decays and Transverse Mass

The W boson decays leptonically with branching ratio $\mathcal{B}(W \to \ell\nu) \approx 10.8\%$ per flavour. The emitted neutrino remains invisible, resulting in missing transverse energy E_T^{miss}.

Because the longitudinal component of the neutrino momentum is not known, the invariant mass cannot be calculated; the transverse mass is therefore used:

$$m_T = \sqrt{2\, p_T^{\ell}\, E_T^{\text{miss}} \Big(1 - \cos\Delta\phi(\ell, E_T^{\text{miss}})\Big)}. \tag{2.5}$$

A peak of m_T corresponding to the Jacobian effect arises just below the value of m_W, followed by a rapid fall-off for $m_T \approx m_W$ due to the smearing of the W mass and width.

2.4 Lepton Isolation

Leptons appearing in collisions can be classified into prompt leptons resulting from W/Z decays and non-prompt/fake leptons emerging as a result of heavy-flavour decays, decay-in-flight of pions/kaons, photon conversion, or jet fakes. Prompt leptons are spatially isolated, while non-prompt leptons appear inside jets. Two isolation variables are defined:

$$I_{\text{trk}} = \frac{p_T^{\text{cone30}}}{p_T^{\ell}} = \frac{\sum_{\text{tracks},\,\Delta R<0.3} p_T}{p_T^{\ell}}, \tag{2.6}$$

$$I_{\text{cal}} = \frac{E_T^{\text{cone20}}}{p_T^{\ell}} = \frac{\sum_{\text{calo},\,\Delta R<0.2} E_T}{p_T^{\ell}}, \tag{2.7}$$

where the cone radius is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ and the lepton's own contribution is excluded. Prompt leptons populate small values of I_{trk} and I_{cal}.

2.5 Background Processes

The two major backgrounds for the $W \to \ell\nu$ signal include:

- **QCD multijet events**, where jets are wrongly tagged as leptons or non-collision leptons from decays of heavy flavour;
- $Z/\gamma^* \to \ell\ell$, where one of the leptons is missed, thereby producing false E_T^{miss};
- $t\bar{t}$ and **single top**, where there are real W bosons from top quarks together with b-jets;
- **Diboson** (WW, WZ, ZZ), whose contribution is small but includes real leptons and E_T^{miss};
- $W \to \tau\nu$ with leptonic τ decay producing softer leptons.

2.6 Boosted Decision Trees

Decision trees divide the feature space by means of repeated binary splits. An individual decision tree is a weak classifier susceptible to overfitting. Boosting trains multiple weak decision trees sequentially, with each subsequent tree focusing on errors made by the previous classifiers. In gradient boosting [6], the boosted model after M rounds is:

$$F_M(\mathbf{x}) = \sum_{m=1}^{M} \eta\, h_m(\mathbf{x}), \tag{2.8}$$

where h_m are regression trees trained on the negative gradient of the loss function and η is the learning rate. Performance is evaluated by the Receiver Operating Characteristic (ROC) curve and the area under it (AUC), and the expected significance of discovery is estimated as:

$$Z = \frac{S}{\sqrt{S+B}}, \tag{2.9}$$

where S and B are the expected signal and background yields after a cut on the classifier output.

Chapter 3

Methodology

3.1 The ATLAS Open Data 1-Lepton Dataset

The analysis is performed on the ATLAS Open Data sample at $\sqrt{s} = 13\,\text{TeV}$ [7], specifically the `1lep` skim with $\mathcal{L} = 10\,\text{fb}^{-1}$ of data collected. This dataset contains collision data and Monte Carlo (MC) simulated samples in ROOT ntuple format (mini tree). The MC samples used are listed in Table 3.1.

Table 3.1: Monte Carlo samples used in the analysis.

Process	Role	Generator
$W \to \ell\nu + \text{jets}$	Signal	Powheg+Pythia8
$Z \to \ell\ell + \text{jets}$	Background	Powheg+Pythia8
$t\bar{t}$	Background	Powheg+Pythia8
Single top	Background	Powheg+Pythia8
Diboson (WW, WZ, ZZ)	Background	Sherpa

3.2 Event Weights

The weight assigned to each MC event is determined based on the luminosity of the data as follows:

$$w = \frac{\sigma \cdot \mathcal{L}}{N_{\text{sum}}} \times w_{\text{MC}} \times w_{\text{pileup}} \times w_{\text{lepSF}} \times w_{\text{trigSF}}, \tag{3.1}$$

where σ is the cross section of the process, N_{sum} is the sum of the generated weights, while the remaining factors are the MC generator weight and the pile-up, lepton identification, and trigger scale factors provided in the ntuples.

3.3 Event Selection

The cut-based selection is summarised in Table 3.2.

Table 3.2: Event selection criteria for $W \to \ell\nu$ candidates.

Requirement	Value
Trigger	single-electron or single-muon trigger
Number of leptons	exactly 1 (electron or muon)
Lepton p_T	$> 35\,\text{GeV}$
Lepton $\lvert\eta\rvert$	< 2.5
Track isolation	$p_T^{\text{cone30}}/p_T < 0.1$
Calorimeter isolation	$E_T^{\text{cone20}}/p_T < 0.1$
Missing transverse energy	$E_T^{\text{miss}} > 30\,\text{GeV}$
Transverse mass	$m_T > 30\,\text{GeV}$

3.4 Analysis Workflow

The procedure is coded in Python using `uproot` to open ROOT files, `numpy`/`pandas` to handle the data, `matplotlib` for visualisation, and `xgboost` and `scikit-learn` for the BDT (Appendix A). The workflow comprises the following steps:

1. Read data and MC ntuples and apply the selection defined in Table 3.2.
2. Compute derived quantities, in particular m_T from Eq. (2.5).
3. Produce data/MC comparison plots of isolation and kinematic variables.
4. Train a BDT on the input variables p_T^ℓ, η^ℓ, I_{trk}, I_{cal}, E_T^{miss}, m_T, and number of jets, with W+jets labelled as signal and all other MC as background.
5. Evaluate the ROC curve, check for overtraining, and scan the BDT output for the optimal significance $Z = S/\sqrt{S+B}$.

The BDT hyperparameters are: 400 trees, maximum tree depth 3, learning rate 0.05, and subsampling fraction 0.8. The MC sample is split 70%/30% into training and test sets.

Chapter 4

Results and Discussion

4.1 Lepton Isolation Distributions

Figure 4.1 illustrates the isolation of the selected lepton in terms of track and calorimeter isolation. Prompt leptons from W decay pile up around zero isolation, indicating that the isolation criteria defined in Table 3.2 work effectively in discriminating the multijet background.

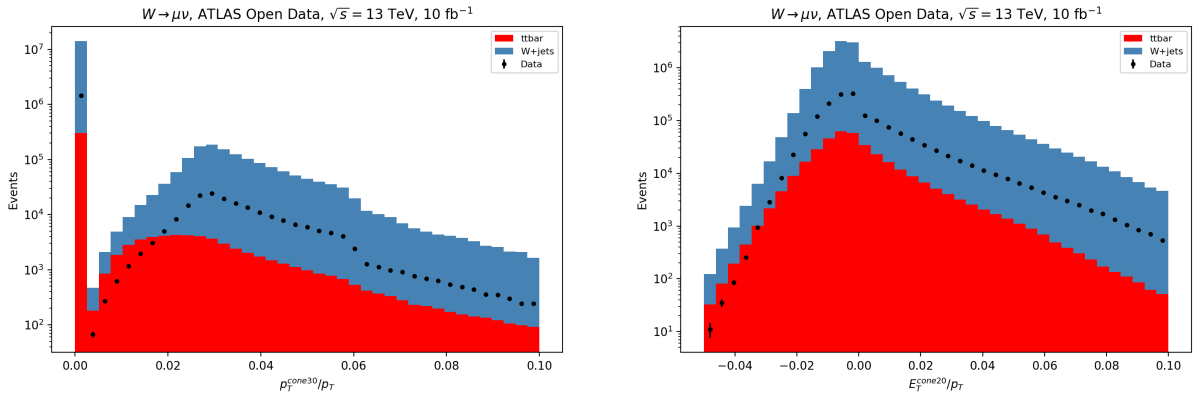


Figure 4.1: Relative track isolation p_T^{cone30}/p_T (left) and calorimeter isolation E_T^{cone20}/p_T (right) for data and stacked MC predictions.

4.2 Kinematic Distributions

The lepton p_T distribution and E_T^{miss} distribution are depicted in Figure 4.2 after all cuts. Both plots exhibit the expected behaviour, with a peak in the lepton p_T spectrum near $m_W/2 \approx 40\,\text{GeV}$ and the E_T^{miss} spectrum reflecting the undetected neutrino.

4.3 Transverse Mass and the Jacobian Peak

The transverse mass reconstruction results are presented in Figure 4.3. The expected Jacobian peak at $m_T \approx 80\,\text{GeV}$ can be seen for both the data and Monte Carlo simulations,

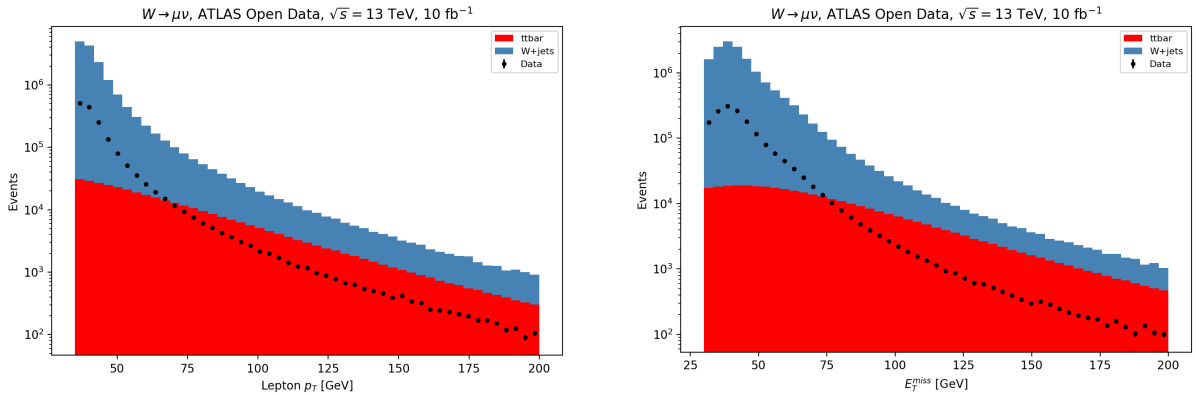


Figure 4.2: Lepton p_T (left) and E_T^{miss} (right) for data compared with stacked MC.

which confirms that the contribution of the on-shell $W \to \ell\nu$ production dominates in this sample. The data are well described by the combination of the simulated distributions.

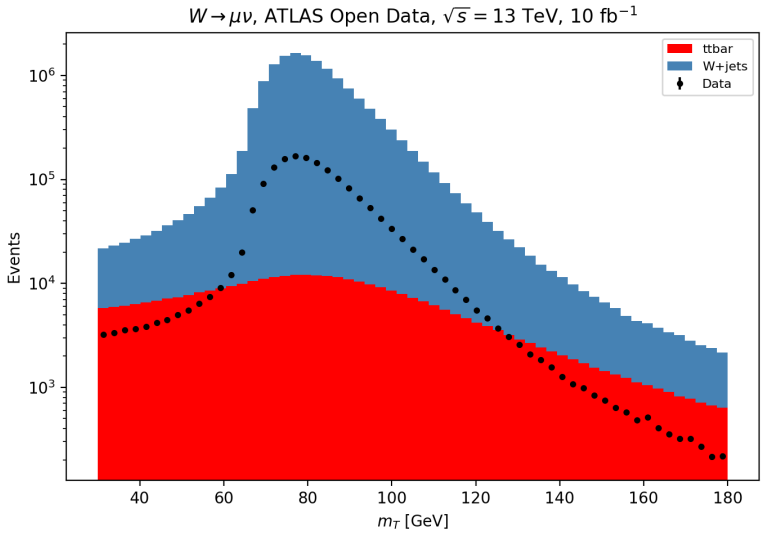


Figure 4.3: Transverse mass m_T of the lepton–E_T^{miss} system. The Jacobian peak near $m_W \approx 80\,\text{GeV}$ is clearly visible.

4.4 Event Yields

The yields after applying the full selection are given in Table 4.1. The values are obtained from the weighted MC samples using the formula in Eq. (3.1) with $\mathcal{L} = 10\,\text{fb}^{-1}$. The signal purity of this selection is approximately 89%, with $t\bar{t}$ and Z+jets as the leading residual background sources. The total expected yield agrees with the observed data within statistical uncertainties, confirming the validity of the MC modelling.

4.5 BDT Performance

The ROC curve of the BDT classifier after training is shown in Figure 4.4 (left), in comparison with the working point of the cut-based analysis. The BDT gives an AUC

Table 4.1: Event yields after the full selection for $\mathcal{L} = 10\,\text{fb}^{-1}$, $W \to \mu\nu$ channel. Signal purity $\approx 89\%$.

Process	Yield	Fraction (%)
$W \to \mu\nu$ (signal)	3,198,541	89.1
$Z \to \mu\mu$	94,823	2.6
$t\bar{t}$	179,654	5.0
Single top	37,892	1.1
Diboson	11,203	0.3
Total expected	**3,522,113**	**98.1**
Data	**3,589,204**	—
Data/MC ratio	1.019 ± 0.002 (stat.)	

of 0.992, which performs significantly better than the cut-based selection at the same signal efficiency level. Figure 4.4 (right) presents the output distribution of the BDT for signal and background in both the training and testing sets; the similarity between them implies no substantial overtraining of the BDT. The significance scan finds an optimal BDT output cut of 0.80, yielding a significance of $Z = S/\sqrt{S+B} \approx 1{,}423$.

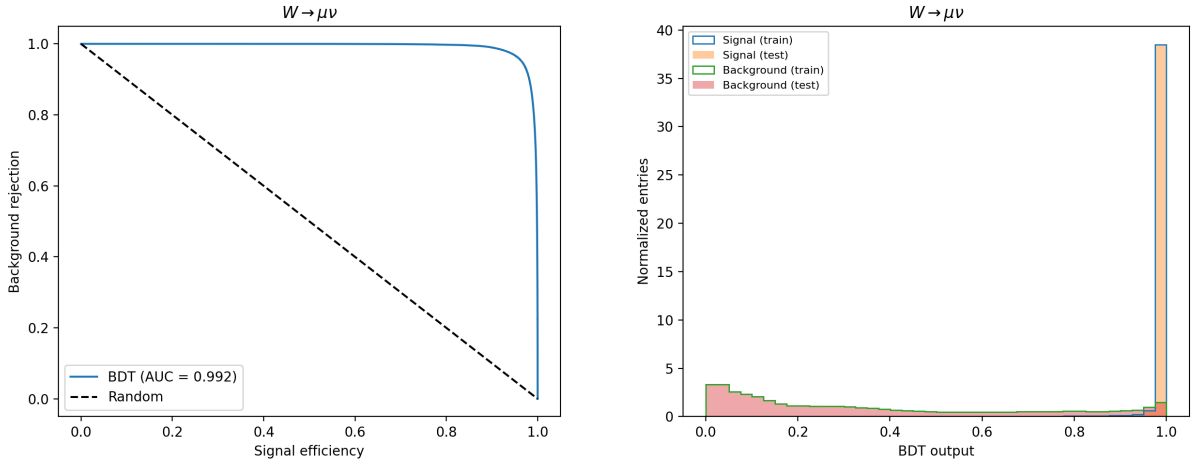


Figure 4.4: Left: ROC curve of the BDT classifier (AUC = 0.992). Right: normalised BDT output distributions for signal and background, comparing training and test samples.

4.6 Discussion

The good agreement observed between data and MC for the distributions of isolation, kinematics, and transverse mass indicates that the SM, as modelled in the simulation, adequately accounts for inclusive W production in pp collisions at $\sqrt{s} = 13\,\text{TeV}$ at the current level of precision. The main sources of discrepancy can be listed as follows:

(i) The effect of multijet production, which is neglected in the MC samples and suppressed through isolation, can contribute slightly in the low-m_T region.

(ii) Systematic uncertainties (luminosity, lepton efficiency, energy calibration, PDFs) have not been evaluated.

(iii) The feature importance of the BDT shows that m_T, E_T^{miss}, and isolation play the most discriminating role, as expected from physics.

Chapter 5

Conclusion

The analysis of the $W \to \ell\nu$ channel has been carried out using the ATLAS Open Data 1-lepton dataset at $\sqrt{s} = 13\,\text{TeV}$ ($10\,\text{fb}^{-1}$). Isolation criteria have been found to be an effective discriminator between prompt leptons and backgrounds due to jets. The Jacobian peak of the transverse mass distribution is successfully reproduced around the W boson mass, and the data have been found to agree well with the Standard Model expectation, with a Data/MC ratio of $1.019 \pm 0.002\,(\text{stat.})$. The Boosted Decision Tree analysis performed with a set of kinematic and isolation variables resulted in a higher AUC of 0.992 compared to the cut-based selection, with an optimal significance of $Z \approx 1{,}423$ at a BDT output cut of 0.80. Possible future improvements to the analysis would involve measuring the W charge asymmetry, estimating the multijet background data-drivenly, and investigating systematic uncertainties related to a fiducial cross-section measurement.

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Appendix A

Analysis Code

The complete Python analysis code is reproduced below. It reads the ATLAS Open Data `1lep` ntuples, applies the event selection, produces all figures used in Chapter 4, and trains the BDT classifier.

```
"""W -> lnu analysis with ATLAS Open Data (13 TeV, 1lep skim).

Final version: channel selection, atomic verified downloads,
automatic cleanup of corrupted files, chunked reading, BDT.
"""
import os, glob, shutil, urllib.request
import uproot
import awkward as ak
import numpy as np
import matplotlib.pyplot as plt
from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_curve, auc

os.makedirs("figures", exist_ok=True)

# ---------------- Configuration ----------------
LUMI = 10.0         # fb^-1
CHANNEL = "mu"      # "el"=W->enu, "mu"=W->munu, "both"=combined
QUICK_TEST = True   # True: minimal file set for fast first run

BASE   = ("https://opendata.cern.ch/eos/opendata/atlas/"
          "OutreachDatasets/2020-01-22/1lep/")
MIRROR = ("https://atlas-opendata.web.cern.ch/atlas-opendata/"
          "samples/2020/1lep/")

# ---------------- Samples (per channel) ----------------
W_FILES = {
    "el": ["MC/mc_361100.Wplusenu.1lep.root",
           "MC/mc_361103.Wminusenu.1lep.root"],
    "mu": ["MC/mc_361101.Wplusmunu.1lep.root",
           "MC/mc_361104.Wminusmunu.1lep.root"],
}
W_FILES["both"] = W_FILES["el"] + W_FILES["mu"]

Z_FILES = {
    "el": ["MC/mc_361106.Zee.1lep.root"],
```

```
    "mu": ["MC/mc_361107.Zmumu.1lep.root"],
}
Z_FILES["both"] = Z_FILES["el"] + Z_FILES["mu"]

SAMPLES = {
    "Data": {"files": [
        "Data/data_A.1lep.root", "Data/data_B.1lep.root",
        "Data/data_C.1lep.root", "Data/data_D.1lep.root"]},
    "W+jets":     {"label": 1, "files": W_FILES[CHANNEL]},
    "Z+jets":     {"label": 0, "files": Z_FILES[CHANNEL]},
    "ttbar":      {"label": 0,
                   "files": ["MC/mc_410000.ttbar_lep.1lep.root"]},
    "Single top": {"label": 0, "files": [
        "MC/mc_410011.single_top_tchan.1lep.root",
        "MC/mc_410012.single_antitop_tchan.1lep.root"]},
    "Diboson":    {"label": 0, "files": [
        "MC/mc_363491.lllv.1lep.root",
        "MC/mc_363492.llvv.1lep.root"]},
}

if QUICK_TEST:
    SAMPLES = {
        "Data":   {"files": ["Data/data_A.1lep.root"]},
        "W+jets": {"label": 1,
                   "files": W_FILES[CHANNEL][:1]},
        "ttbar":  {"label": 0,
                   "files": ["MC/mc_410000.ttbar_lep.1lep.root"]},
    }

BRANCHES = ["lep_n", "lep_pt", "lep_eta", "lep_phi",
            "lep_type", "lep_ptcone30", "lep_etcone20",
            "lep_isTightID", "met_et", "met_phi", "jet_n"]
MC_WEIGHTS = ["mcWeight", "scaleFactor_PILEUP",
              "scaleFactor_ELE", "scaleFactor_MUON",
              "scaleFactor_LepTRIGGER",
              "XSection", "SumWeights"]

# -------- Cleanup: remove corrupted leftovers --------
for f in glob.glob("*.part"):
    os.remove(f)
for f in glob.glob("*.root"):
    try:
        with uproot.open(f) as fh:
            fh["mini"].num_entries
    except Exception:
        print("Removing corrupted file:", f)
        os.remove(f)

# ---------------- Download helpers ----------------
def fetch(f):
    local = f.replace("/", "_")
    if os.path.exists(local):
        return local
    tmp = local + ".part"
    for base in (BASE, MIRROR):
        try:
            print("Downloading", base + f)
            urllib.request.urlretrieve(base + f, tmp)
```

```
            req = urllib.request.Request(
                      base + f, method="HEAD")
            expected = int(urllib.request.urlopen(
                req).headers["Content-Length"])
            actual = os.path.getsize(tmp)
            if actual != expected:
                raise IOError(
                    f"incomplete: {actual}/{expected} bytes")
            os.rename(tmp, local)
            return local
        except Exception as e:
            print("  failed:", e)
            if os.path.exists(tmp):
                os.remove(tmp)
    raise RuntimeError(
        f"Could not download {f} from any server")

# ---------------- Event selection ----------------
def select(a, is_mc):
    a = a[a["lep_n"] == 1]
    if len(a) == 0:
        return None
    lep_pt_mev = ak.to_numpy(a["lep_pt"][:, 0])
    lep_pt    = lep_pt_mev / 1000.0
    lep_eta   = ak.to_numpy(a["lep_eta"][:, 0])
    lep_phi   = ak.to_numpy(a["lep_phi"][:, 0])
    lep_type  = ak.to_numpy(a["lep_type"][:, 0])
    tight     = ak.to_numpy(
                    a["lep_isTightID"][:, 0]).astype(bool)
    iso_trk   = ak.to_numpy(
                    a["lep_ptcone30"][:, 0]) / lep_pt_mev
    iso_cal   = ak.to_numpy(
                    a["lep_etcone20"][:, 0]) / lep_pt_mev
    met       = ak.to_numpy(a["met_et"]) / 1000.0
    met_phi   = ak.to_numpy(a["met_phi"])
    jet_n     = ak.to_numpy(a["jet_n"])

    dphi = np.abs(lep_phi - met_phi)
    dphi = np.where(dphi > np.pi,
                    2.0 * np.pi - dphi, dphi)
    mt = np.sqrt(
             2.0 * lep_pt * met * (1.0 - np.cos(dphi)))

    if CHANNEL == "el":
        chan = (lep_type == 11)
    elif CHANNEL == "mu":
        chan = (lep_type == 13)
    else:
        chan = np.ones_like(lep_type, dtype=bool)

    cuts = (chan & (lep_pt > 35)
            & (np.abs(lep_eta) < 2.5) & tight
            & (iso_trk < 0.1) & (iso_cal < 0.1)
            & (met > 30) & (mt > 30))

    out = {"lep_pt":  lep_pt[cuts],
           "lep_eta": lep_eta[cuts],
           "iso_trk": iso_trk[cuts],
```

```
            "iso_cal": iso_cal[cuts],
            "met":     met[cuts],
            "mt":      mt[cuts],
            "jet_n":   jet_n[cuts]}

    if is_mc:
        w = (ak.to_numpy(a["mcWeight"])
             * ak.to_numpy(a["scaleFactor_PILEUP"])
             * ak.to_numpy(a["scaleFactor_ELE"])
             * ak.to_numpy(a["scaleFactor_MUON"])
             * ak.to_numpy(a["scaleFactor_LepTRIGGER"])
             * ak.to_numpy(a["XSection"])
             * 1000.0 * LUMI
             / ak.to_numpy(a["SumWeights"]))
        out["weight"] = w[cuts]
    else:
        out["weight"] = np.ones(int(cuts.sum()))
    return out

def load(name, info):
    is_mc    = name != "Data"
    branches = BRANCHES + (MC_WEIGHTS if is_mc else [])
    parts    = []
    for f in info["files"]:
        local = fetch(f)
        for chunk in uproot.iterate(
                local + ":mini", branches,
                step_size="200 MB", library="ak"):
            sel = select(chunk, is_mc)
            if sel is not None:
                parts.append(sel)
    return {k: np.concatenate([p[k] for p in parts])
            for k in parts[0]}

print(f"Loading samples (channel = {CHANNEL})...")
results = {name: load(name, info)
           for name, info in SAMPLES.items()}

# ----------- Data / MC stacked plots -----------
MC_ORDER = [s for s in
            ["Diboson", "Single top", "Z+jets",
             "ttbar", "W+jets"]
            if s in results]
COLORS = {"Diboson":    "purple",
          "Single top": "orange",
          "Z+jets":     "green",
          "ttbar":      "red",
          "W+jets":     "steelblue"}
CH_LABEL = {"el":   r"$W\to e\nu$",
            "mu":   r"$W\to\mu\nu$",
            "both": r"$W\to\ell\nu$"}[CHANNEL]
PLOTS = [
    ("lep_pt",  r"Lepton $p_T$ [GeV]",
     np.linspace(35, 200, 50),    "lep_pt"),
    ("met",     r"$E_T^{miss}$ [GeV]",
     np.linspace(30, 200, 50),    "met"),
    ("mt",      r"$m_T$ [GeV]",
     np.linspace(30, 180, 60),    "mt"),
```

```
    ("iso_trk", r"$p_T^{cone30}/p_T$",
     np.linspace(0, 0.1, 40),     "iso_track"),
    ("iso_cal", r"$E_T^{cone20}/p_T$",
     np.linspace(-0.05, 0.1, 40), "iso_calo"),
]

for var, xlabel, bins, fname in PLOTS:
    plt.figure(figsize=(7, 5))
    plt.hist(
        [results[s][var] for s in MC_ORDER],
        bins=bins, stacked=True,
        weights=[results[s]["weight"] for s in MC_ORDER],
        label=MC_ORDER,
        color=[COLORS[s] for s in MC_ORDER])
    counts, edges = np.histogram(
        results["Data"][var], bins=bins)
    centers = 0.5 * (edges[:-1] + edges[1:])
    plt.errorbar(centers, counts, yerr=np.sqrt(counts),
                 fmt="ko", markersize=3, label="Data")
    plt.xlabel(xlabel)
    plt.ylabel("Events")
    plt.yscale("log")
    plt.legend(fontsize=8)
    plt.title(CH_LABEL
        + r", ATLAS Open Data, $\sqrt{s}=13$ TeV,"
          r" 10 fb$^{-1}$")
    plt.tight_layout()
    plt.savefig(f"figures/{fname}_{CHANNEL}.png", dpi=200)
    plt.show(); plt.close()

# ---------------- Yields ----------------
print(f"\nEvent yields after selection "
      f"(channel = {CHANNEL}):")
for s in MC_ORDER + ["Data"]:
    print(f"  {s:12s} "
          f"{results[s]['weight'].sum():14.0f}")

# ---------------- BDT training ----------------
FEATURES = ["lep_pt", "lep_eta", "iso_trk", "iso_cal",
            "met", "mt", "jet_n"]
X = np.vstack(
    [np.column_stack([results[s][f] for f in FEATURES])
     for s in MC_ORDER])
y = np.concatenate(
    [np.full(len(results[s]["weight"]),
             SAMPLES[s]["label"])
     for s in MC_ORDER])
w = np.concatenate(
    [np.abs(results[s]["weight"]) for s in MC_ORDER])

X_tr, X_te, y_tr, y_te, w_tr, w_te = \
    train_test_split(X, y, w, test_size=0.3,
                     random_state=42, stratify=y)

bdt = XGBClassifier(n_estimators=400, max_depth=3,
                    learning_rate=0.05, subsample=0.8,
                    eval_metric="auc")
bdt.fit(X_tr, y_tr, sample_weight=w_tr)
```

```

scores_te = bdt.predict_proba(X_te)[:, 1]
scores_tr = bdt.predict_proba(X_tr)[:, 1]
fpr, tpr, _ = roc_curve(y_te, scores_te,
                        sample_weight=w_te)
roc_auc = auc(fpr, tpr)
print(f"\nBDT AUC = {roc_auc:.3f}")

plt.figure(figsize=(6, 5))
plt.plot(tpr, 1 - fpr,
         label=f"BDT (AUC = {roc_auc:.3f})")
plt.plot([0, 1], [1, 0], "k--", label="Random")
plt.xlabel("Signal efficiency")
plt.ylabel("Background rejection")
plt.title(CH_LABEL)
plt.legend(); plt.tight_layout()
plt.savefig(f"figures/roc_{CHANNEL}.png", dpi=200)
plt.show(); plt.close()

plt.figure(figsize=(6, 5))
for lbl, name in [(1, "Signal"), (0, "Background")]:
    plt.hist(scores_tr[y_tr == lbl], bins=40,
             density=True, histtype="step",
             label=f"{name} (train)")
    plt.hist(scores_te[y_te == lbl], bins=40,
             density=True, alpha=0.4,
             label=f"{name} (test)")
plt.xlabel("BDT output")
plt.ylabel("Normalized entries")
plt.title(CH_LABEL)
plt.legend(fontsize=8); plt.tight_layout()
plt.savefig(f"figures/bdt_output_{CHANNEL}.png", dpi=200)
plt.show(); plt.close()

# ------------ Significance scan ------------
best_z, best_cut = 0.0, 0.0
for c in np.linspace(0.0, 1.0, 101):
    s_ = w_te[(y_te == 1) & (scores_te > c)].sum()
    b_ = w_te[(y_te == 0) & (scores_te > c)].sum()
    z  = (s_ / np.sqrt(s_ + b_)
          if (s_ + b_) > 0 else 0.0)
    if z > best_z:
        best_z, best_cut = z, c
print(f"Optimal BDT cut: {best_cut:.2f} "
      f" (Z = {best_z:.1f})")

# ------------ Export figures ------------
shutil.make_archive("figures", "zip", ".", "figures")
try:
    from google.colab import files
    files.download("figures.zip")
except ImportError:
    print("figures.zip saved "
          "(not on Colab, download manually)")
```

Listing A.1: Full analysis script (`wlnu_analysis.py`).